

SAHIL KUMAR

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EDUCATION

Indian Institute of Information Technology, Kalyani

B.Tech - Computer Science and Engineering - **8.27 CGPA** - **6th Semester**

Nov 2022 – July 2026

Kalyani, India

EXPERIENCE

3W Business Private Limited

Nov 2024 – Feb 2025

Backend Intern ↗

- Engineered a **real-time notification system** using Socket.io, increasing user engagement.
- Architected a **role-based authorization framework** supporting moderator lifecycle operations, including user conversion and privilege management.
- Devised and deployed a **bulk email system** enabling efficient mass communication.

PROJECTS

Autism Detection ↗ | Python, Scikit-learn, NumPy, Pandas

Live

- Designed an autism detection system leveraging **HOG-based** facial features; trained and evaluated **KNN, SVM, XGBoost, and Random Forest** models.
- Achieved 90%+ precision and recall on 130+ facial images using **XGBoost** and **Random Forest**.
- Launched a Streamlit app for real-time image upload and prediction (<5s inference).

E-Commerce Backend ↗ | Node.js, Express.js, REST APIs, MongoDB

Backend

- Developed RESTful APIs for **core e-commerce flows (cart, wishlist, orders)**, reducing response time by 35% and boosting user engagement.
- Optimized MongoDB queries with validation and indexing, improving performance by 30%.
- Implemented **JWT-based authentication** for secure, scalable user access control.

Chat App ↗ | MERN Stack, Socket.io, JWT

Backend Frontend

- Built a real-time chat app using MERN, with **REST APIs, Socket.io, and JWT** for secure, **low-latency 1:1 and chatroom messaging**.
- Integrated features like **online status, typing indicators, read receipts, and message deletion** to enhance interactivity and user control.
- Applied the **useContext** hook for efficient state handling and implemented **responsive design** principles for cross-device UX, supported by a scalable **MVC backend**.

TECHNICAL SKILLS

Languages: C, C++, JavaScript, Python, SQL

Developer Tools: VS Code, Git, GitHub, Postman, Jupyter Notebook

Web Development: HTML, CSS, React.js, React-Redux, Node.js, Express.js, RESTful APIs, MongoDB, Authentication and Authorization

Frameworks and Libraries: Socket.io, Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn

COURSEWORK / SKILLS

- Data Structures and Algorithms
- Operating Systems
- OOP Concepts
- Machine Learning
- Computer Networks
- Database Management Systems
- C/C++
- Software Engineering